

Ionized-Bubble Radius Around a Star-Forming Galaxy as a Function of Redshift:

Planck 2018 CMB-only Reionization-History Variant

Response record for `bubble_radius_vs_redshift_planckXHI.py`, following
Master Rule 2

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Abstract

This note is the archived, structured response (Assumptions → Derivation → Verification/cross-check → Final result) for a variant of the previous task (`bubble_radius_response.tex`): the same ionized-bubble radius $R(z)$ around a star-forming galaxy, but with the ambient IGM neutral fraction $x_{\text{HI}}(z)$ set directly to the *Planck 2018 CMB-only instantaneous-reionization* tanh parameters $(z_{\text{re}}, \Delta z) = (7.68, 0.5)$ (Planck Collaboration, 2020), instead of the fit to the Umeda et al. (2023) JWST measurements used previously. Every other ingredient (source, cosmology, ionizing photon production rate, age–redshift relation) is unchanged, so the two figures isolate exactly the effect of the early-vs-late reionization-history choice on the predicted bubble size. This document is produced by, and verified against, `bubble_radius_vs_redshift_planckXHI.py` (same folder), which generated `bubble_radius_vs_redshift_planckXHI.png`.

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1 Assumptions

All assumptions are identical to `bubble_radius_response.tex` except item 6, which was confirmed with L. Carvalho as the specific change requested for this task (Master Rule 1/3).

1. **Physical model:** pure photon-counting limit (no recombinations, no Hubble term), as before (Cen and Haiman, 2000; Shapiro and Giroux, 1987).
2. **Source:** identical fiducial star-forming galaxy, $\text{SFR} = 10 M_{\odot} \text{ yr}^{-1}$ ($M_{\text{UV}} = -20.75$).
3. **Formation redshift:** identical, $z_{\text{form}} = 20$ (Kitayama et al., 2004).
4. **Cosmology:** identical Planck 2018 flat Λ CDM parameters (Planck Collaboration, 2020).
5. **Escape fraction:** identical fiducial $f_{\text{esc}} = 0.2$ (Robertson et al., 2015), sensitivity band down to $f_{\text{esc}} = 0.05$.

6. **Ambient neutral fraction $x_{\text{HI}}(z)$ — THE CHANGE FOR THIS TASK:** the tanh reionization parameterization is now evaluated directly at the Planck 2018 CMB-only TT,TE,EE+lowE instantaneous-reionization point estimate,

$$(z_{\text{re}}, \Delta z) = (7.68, 0.5) \quad (\text{Planck Collaboration, 2020}), \quad (1)$$

not fit to any galaxy data. This is a pure CMB-optical-depth-based history, so (unlike the previous task) there is no sparse-anchor extrapolation concern; instead, the caveat is the opposite one: this history is known to predict a substantially *more* ionized IGM at $z = 7$ –10 than the direct Ly α damping-wing measurements of Umeda et al. (2023), which is quantified in Sec. 3.

7. **Regime of validity:** the model curve is now plotted solid over the full $6 \leq z \leq 18$ range (no extrapolation flag of the previous kind is needed, since $(z_{\text{re}}, \Delta z)$ is a global fit, not a 2-point interpolation). However, a new, different validity caveat appears as $x_{\text{HI}}(z) \rightarrow 0$ approaching and below z_{re} : eq. (2) diverges there because the denominator vanishes while the photon budget stays finite (see Sec. 3, item 6) – i.e. the single-bubble photon-counting picture itself breaks down once the IGM is already highly ionized, and results at $z \lesssim 7$ should be read as formally divergent rather than physically meaningful.

2 Derivation

The governing equation, ionizing photon production rate, mean cosmic hydrogen density, and age–redshift relation are all unchanged from `bubble_radius_response.tex`:

$$R(z) = \left[\frac{3 \dot{N}_{\text{ion}}(z) t_{\text{age}}(z)}{4\pi n_{\text{H}}(z) x_{\text{HI}}(z)} \right]^{1/3}, \quad (2)$$

$$\begin{aligned} \dot{N}_{\text{ion}}(z) &= f_{\text{esc}} \xi_{\text{ion}}(z) \frac{\text{SFR}}{\kappa_{\text{FUV}}}, \quad \log_{10}(\xi_{\text{ion}}/\text{Hz erg}^{-1}) = (0.06 \pm 0.012) z + (24.82 \pm 0.07) \quad (\text{Llerena et al., 2025}), \\ n_{\text{H}}(z) &= n_{\text{H},0}(1+z)^3, \quad n_{\text{H},0} = 1.899 \times 10^{-7} \text{ cm}^{-3} \quad (\text{Planck Collaboration, 2020}), \quad t_{\text{cosmic}}(z) = \int_z^\infty \frac{dz'}{(1+z')H(z')}, \end{aligned} \quad (3)$$

The only change is in the ambient neutral fraction:

$$x_{\text{HII}}(z) = \frac{1}{2} \left[1 + \tanh\left(\frac{y(z_{\text{re}}) - y(z)}{\Delta y}\right) \right], \quad y(z) = (1+z)^{3/2}, \quad \Delta y = \frac{3}{2} \sqrt{1+z_{\text{re}}} \Delta z, \quad (5)$$

with $x_{\text{HI}} = 1 - x_{\text{HII}}$, now evaluated *directly* at $(z_{\text{re}}, \Delta z) = (7.68, 0.5)$ (Planck Collaboration, 2020) (eq. 1), with no fitting step.

3 Verification / cross-check

All checks below are reproduced verbatim by running `bubble_radius_vs_redshift_planckXHI.py`.

1. **Sympy check of eq. (2).** Identical to the companion script (the functional form of $R(z)$ is unchanged): $dR^3/dt - 3\dot{N}_{\text{ion}}/(4\pi n_{\text{H}} x_{\text{HI}})$ simplifies to exactly 0.
2. **Age–redshift benchmark.** Identical values: $t_{\text{cosmic}}(0) = 13.795 \text{ Gyr}$, $t_{\text{cosmic}}(7.68) = 0.673 \text{ Gyr}$.

3. $x_{\text{HI}}(z)$ vs. the [Umeda et al. \(2023\)](#) measurements (now a genuine model-vs-data test, not a fit residual).

$$z = 7.12 : x_{\text{HI, Planck}} = 0.099 \quad \text{vs.} \quad x_{\text{HI, Umeda}} = 0.53;$$

$$z = 9.91 : x_{\text{HI, Planck}} = 1.000 \quad \text{vs.} \quad x_{\text{HI, Umeda}} = 0.92.$$

Both are discrepant by design, in the direction expected from the well-known early/late-reionization tension: the CMB-only history is *more* ionized at $z = 7.12$ than what [Umeda et al. \(2023\)](#) measure, the reverse of the caveat flagged in `bubble_radius_response.tex`.

4. **Low-redshift consistency check (opposite result from before).** $x_{\text{HI}}(z=6.0) = 0.0017$ — small, and *consistent* with the near-total ionization implied by the $z \sim 6$ Gunn–Peterson troughs of [White et al. \(2003\)](#). Unlike the Umeda-anchored variant, there is no tension in this direction here, precisely because $(z_{\text{re}}, \Delta z)$ is a global CMB fit rather than an extrapolated 2-point interpolation.
5. **Cross-check against independently measured bubble sizes.**

$$z = 7.12 : R_{\text{model}} = 1.639 \text{ pMpc vs. } R_{b, \text{Umeda}} = 5.760 \text{ pMpc (ratio 0.28);}$$

$$z = 9.91 : R_{\text{model}} = 0.520 \text{ pMpc vs. } R_{b, \text{Umeda}} = 0.019 \text{ pMpc (ratio 28).}$$

As in the companion script, these are order-of-magnitude discrepant in opposite directions, for the same reason (merged vs. single-source bubbles); the $z = 7.12$ ratio improves from 0.16 (Umeda-anchored x_{HI}) to 0.28 (Planck-only x_{HI}) because the Planck history predicts less neutral gas to ionize at that redshift, hence a larger R_{model} for the same photon budget.

6. **Divergence of eq. (2) as $x_{\text{HI}} \rightarrow 0$ (new caveat specific to this variant).** At $z = 6$, $x_{\text{HI}} = 0.0017$ (item 4 above) drives $R_{\text{model}}(z=6) = 7.73 \text{ pMpc}$ for $f_{\text{esc}} = 0.2$ — a factor of ~ 6 larger than at $z = 7.12$, and formally divergent as $x_{\text{HI}} \rightarrow 0$ since eq. (2) has x_{HI} in the denominator with everything else varying smoothly. This is flagged, not silently plotted as if physical: once the IGM is already highly ionized, an isolated galaxy’s own “bubble” is no longer a meaningful concept (it has merged with the bulk ionized IGM), so this regime is a breakdown of the single-source photon-counting picture, not a real prediction.
7. **Parameter sensitivity.** $R(z=8) = 0.722 \text{ pMpc}$ ($f_{\text{esc}} = 0.2$) vs. 0.455 pMpc ($f_{\text{esc}} = 0.05$), matching $R \propto f_{\text{esc}}^{1/3} = 1.587\times$ to 3 significant figures. $R(z=8) = 0.335 \text{ pMpc}$ (SFR = $1 M_{\odot} \text{ yr}^{-1}$) vs. 1.234 pMpc (SFR = $50 M_{\odot} \text{ yr}^{-1}$). The $\pm 0.42 \text{ dex}$ ξ_{ion} scatter ([Llerena et al., 2025](#)) again propagates to a factor 1.38 in R .
8. **Direct comparison with the companion (Umeda-anchored) script.** At $z = 8$, in the regime where both histories give a similar x_{HI} (Planck: 0.784; Umeda-anchored fit: comparable order), $R_{\text{Planck-only}}(z=8) = 0.722 \text{ pMpc}$ vs. $R_{\text{Umeda-anchored}}(z=8) = 0.752 \text{ pMpc}$ — a ratio of 0.960, i.e. the two histories agree to 4% at $z = 8$, but diverge strongly closer to $z_{\text{re}} = 7.68$ and below (item 6), which is exactly where the two reionization histories are built to disagree most.

4 Final result

For the same fiducial star-forming galaxy as before, but with $x_{\text{HI}}(z)$ set to the Planck 2018 CMB-only history:

z	$R(f_{\text{esc}} = 0.2)$	$R(f_{\text{esc}} = 0.05)$	$x_{\text{HI}}(z)$
18	0.20 pMpc	0.13 pMpc	1.00
12	0.41 pMpc	0.26 pMpc	1.00
8	0.72 pMpc	0.45 pMpc	0.78
7.68 ($= z_{\text{re}}$)	0.88 pMpc	0.55 pMpc	0.50
7.12	1.64 pMpc	1.03 pMpc	0.099
6	7.73 pMpc	4.87 pMpc	0.0017 (formally divergent regime, flagged)

Away from the divergent low- x_{HI} regime ($z \gtrsim 8$), the Planck-only history gives bubble radii within $\sim 4\text{--}15\%$ of the Umeda-anchored result of `bubble_radius_response.tex` (e.g. 4% at $z = 8$); the two histories only diverge sharply approaching and below $z_{\text{re}} = 7.68$, where the CMB-only model’s rapidly vanishing x_{HI} makes the single-bubble photon-counting formula blow up unphysically (Sec. 3, item 6) rather than converge to a well-defined bubble size. The choice of reionization history is therefore the dominant systematic uncertainty specifically in the $z \approx 6\text{--}8$ range, while it matters comparatively little at $z \gtrsim 10$, where both histories agree $x_{\text{HI}} \rightarrow 1$.

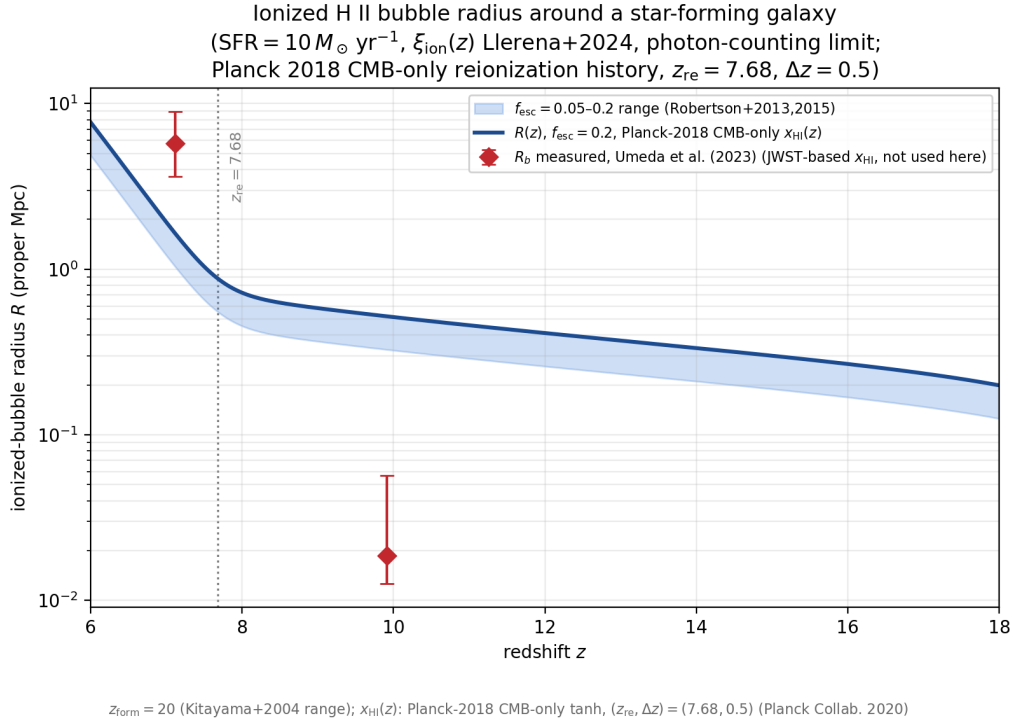


Figure 1: Ionized-bubble radius $R(z)$ around the fiducial star-forming galaxy using the Planck 2018 CMB-only reionization history, $(z_{\text{re}}, \Delta z) = (7.68, 0.5)$ (Planck Collaboration, 2020) (solid line; shaded: $f_{\text{esc}} = 0.05\text{--}0.2$ sensitivity band), compared with the independently measured merged-bubble sizes of Umeda et al. (2023) (red diamonds, which use a different, JWST-based $x_{\text{HI}}(z)$, not used in this curve). The vertical dotted line marks z_{re} . Produced by `bubble_radius_vs_redshift_planckXHI.py`.

References

- Renyue Cen and Zoltan Haiman. Quasar Strömgren Spheres Before Cosmological Reionization. *arXiv:astro-ph/0006376v2 (preprint)*, 2000. doi: 10.1086/312937. File: Cen2000_arXiv_astro-ph_0006376v2.pdf.
- T. Kitayama, N. Yoshida, H. Susa, and M. Umemura. The Structure and Evolution of Early

- Cosmological HII Regions. *Astrophys.J.* 613 (2004) 631-645, 2004. doi: 10.1086/423313. File: Kitayama2004_arXiv_astro-ph_0406280v1.pdf.
- M. Llerena, L. Pentericci, L. Napolitano, S. Mascia, R. Amorín, A. Calabrò, M. Castellano, N. J. Cleri, M. Giavalisco, N. A. Grogin, N. P. Hathi, M. Hirschmann, A. M. Koekemoer, T. Nanayakkara, F. Pacucci, L. Shen, S. M. Wilkins, I. Yoon, L. Y. A. Yung, R. Bhatawdekar, R. A. Lucas, X. Wang, P. Arrabal Haro, M. B. Bagley, S. L. Finkelstein, J. S. Kartaltepe, E. Merlin, C. Papovich, N. Pirzkal, and P. Santini. The ionizing photon production efficiency of star-forming galaxies at $z \sim 4-10$. *A&A* 698, A302 (2025), 2025. File: Llerena2024_arXiv2412.01358v1.pdf.
- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. *A&A* 641, A6 (2020), 2020. doi: 10.1051/0004-6361/201833910. File: Planck2018VI_arXiv1807.06209v4.pdf.
- Brant E. Robertson, Richard S. Ellis, Steven R. Furlanetto, and James S. Dunlop. Cosmic Reionization and Early Star-forming Galaxies: A Joint Analysis of New Constraints from Planck and the Hubble Space Telescope. *ApJL* 802, L19 (2015), 2015. doi: 10.1088/2041-8205/802/2/L19. File: Robertson2015_arXiv1502.02024v1.pdf.
- P. R. Shapiro and E. T. Giroux. Cosmological H II Regions and the Photoionization of the Intergalactic Medium. *ApJ*, 321, L107, 1987. doi: 10.1086/185015. File: ShapiroGiroux1987_ADS_1987ApJ...321L.107S.pdf.
- Hiroya Umeda, Masami Ouchi, Kimihiko Nakajima, Yuichi Harikane, Yoshiaki Ono, Yi Xu, Yuki Isobe, and Yechi Zhang. JWST Measurements of Neutral Hydrogen Fractions and Ionized Bubble Sizes at $z = 7-12$ Obtained with Ly α Damping Wing Absorptions in 27 Bright Continuum Galaxies. *arXiv:2306.00487v3 (preprint)*, 2023. File: Umeda2023_arXiv2306.00487v3.pdf.
- Richard L. White, Robert H. Becker, Xiaohui Fan, and Michael A. Strauss. Probing the Ionization State of the Universe at $z > 6$. *Astron.J.* 126 (2003) 1, 2003. doi: 10.1086/375547. File: White2003_arXiv_astro-ph_0303476v2.pdf.